

BRANDON TANAKA CHAPWANYA
FE545-A

VS CODE OUTPUT: SCREENSHORTS

```
o tanakasmacbookair@Tanakas-MacBook-Air PART 1 % ./quant_app
```

```
Number of steps  
50
```

```
Number of steps: 50
```

```
Parameters Used:  
Spot: 50  
Strike: 50  
Volatility: 0.3  
Risk-free rate: 0.05  
Dividend yield: 0.08  
Expiry: 1  
Number of steps: 50
```

	American Call	American Put
CRR Binomial Tree	4.004509	8.078126
Boyle Trinomial Tree	3.998631	8.088431

PART 2

```
tanakasmacbookair@Tanakas-MacBook-Air FINAL EXAM % g++ -std=c++11 -o Part2 Part2_Main.cpp CRRBinomialTree.cpp TrinomialTree.cpp TreeEuropeanBarrier.cpp Arrays.cpp Parameters.cpp PayOff3.cpp PayOffBridge.cpp TreeProducts.cpp TreeAmerican.cpp TreeEuropean.cpp BlackScholesFormulas.cpp Normals.cpp && ./Part2
2 warnings generated.
```

```
Number of steps
50
```

```
Parameters Used:
Spot: 50
Strike: 50
Barrier: 70
Volatility: 0.3
Risk-free rate: 0.05
Dividend yield: 0.08
Expiry: 1
Number of steps: 50
```

```
European Knock-Out Barrier Option Prices:
```

	Barrier Call	Barrier Put
CRR Binomial Tree	1.139640	8.034440
Boyle Trinomial Tree	1.179251	8.049692

PART 3

```
tanakasmacbookair@Tanakas-MacBook-Air FINAL EXAM % g++ -std=c++11 -o Part3 Part3.Main.cpp CRRBinomialTree.cpp TrinomialTree.cpp TreeAmerican.cpp TreeEuropeanBarrier.cpp  
rays.cpp Parameters.cpp PayOff3.cpp PayOffBridge.cpp TreeProducts.cpp TreeEuropean.cpp BlackScholesFormulas.cpp Normals.cpp && ./Part3
```

```
Arrays.cpp:265:8: warning: extra tokens at end of #endif directive [-Wextra-tokens]
 265 | #endif RANGE_CHECKING
    | ^
2 warnings generated.
```

Parameters Used:
Spot: 50
Strike: 50
Barrier: 70
Volatility: 0.3
Risk-free rate: 0.05
Dividend yield: 0.08
Expiry: 1

Comparing American Call and European Knock-Out Barrier Call
across different numbers of time steps:

Steps (N)	American Call (CRR)	Barrier Call (CRR)	American Call (Tri)	Barrier Call (Tri)
50	4.004509	1.139640	3.998631	1.179251
100	4.003734	1.125936	4.004983	1.176280
200	4.006687	1.120027	4.003302	1.140077
400	4.006543	1.098615	4.005348	1.126103

Analysis:

- (a) The knock-out barrier call is worth less than the American call because the barrier feature adds a condition that makes the option worthless if the stock price ever reaches or exceeds $B=70$ during its life. Many paths that would produce a positive payoff for the American call get knocked out and pay zero instead. The American call has no such restriction and also benefits from early exercise.
- (b) As the number of time steps increases from 50 to 400, the option prices converge toward stable values. With more steps, the tree approximates the continuous-time model more accurately, reducing discretization error. The barrier condition also becomes more precise because the tree checks the barrier more frequently, making knock-out detection more accurate along each path.

PART 4

```
tanakasmacbookair@Tanakas-MacBook-Air FINAL EXAM % g++ -std=c++11 -o Part4 Part4_Main.cpp BiTAmeriCall.cpp BiTAmeriPut.cpp BiTEuroBarrierCall.cpp BiTEuroBarrierPut.cpp PayOffConstructible.cpp CRRBinomialTree.cpp TreeAmerican.cpp TreeEuropeanBarrier.cpp Arrays.cpp Parameters.cpp PayOff3.cpp PayOffBridge.cpp TreeProducts.cpp ArgList.cpp BlackScholesFormulas.cpp Normals.cpp && ./Part4
Arrays.cpp:247:8: warning: extra tokens at end of #endif directive [-Wextra-tokens]
 247 | #endif RANGE_CHECKING
      | ^
      | //
Arrays.cpp:265:8: warning: extra tokens at end of #endif directive [-Wextra-tokens]
 265 | #endif RANGE_CHECKING
      | ^
      | //
2 warnings generated.

Number of steps
50

Parameters Used:
Spot: 50
Strike: 50
Barrier: 70
Volatility: 0.3
Risk-free rate: 0.05
Dividend yield: 0.08
Expiry: 1
Number of steps: 50

Option Prices using PayOffFactory:
+-----+-----+
|Option ID|Price |
+-----+-----+
|BiTAmeriCall|4.004509|
+-----+-----+
|BiTAmeriPut|8.078126|
+-----+-----+
|BiTEuroBarrierCall|1.139640|
+-----+-----+
|BiTEuroBarrierPut|8.034440|
+-----+-----+
```

PART 5

```
tanakasmacbookair@Tanakas-MacBook-Air FINAL EXAM % g++ -std=c++11 -o Part5 Part5_Main.cpp TreeObservable.cpp OptionPricingReportWriter.cpp TreePricerWithObserver.cpp CRRBinomialTree.cpp TrinomialTree.cpp TreeAmerican.cpp TreeEuropeanBarrier.cpp Arrays.cpp Parameters.cpp PayOff3.cpp PayOffBridge.cpp TreeProducts.cpp BlackScholesFormulas.cpp Normals.cpp -I/opt/homebrew/include -L/opt/homebrew/lib -lQuantLib && ./Part5

[-Wc++17-extensions]
37 | namespace QuantLib::ext {
    | ~~~~~
    | { namespace ext
1 warning generated.
In file included from TreeObservable.cpp:6:
In file included from ./TreeObservable.h:10:
In file included from /opt/homebrew/include/ql/patterns/observable.hpp:32:
In file included from /opt/homebrew/include/ql/errors.hpp:29:
/opt/homebrew/include/ql/shared_ptr.hpp:37:19: warning: nested namespace definition is a C++17 extension; define each namespace separately
[-Wc++17-extensions]
37 | namespace QuantLib::ext {
    | ~~~~~
    | { namespace ext
1 warning generated.
In file included from OptionPricingReportWriter.cpp:6:
In file included from ./OptionPricingReportWriter.h:9:
In file included from /opt/homebrew/include/ql/patterns/observable.hpp:32:
In file included from /opt/homebrew/include/ql/errors.hpp:29:
/opt/homebrew/include/ql/shared_ptr.hpp:37:19: warning: nested namespace definition is a C++17 extension; define each namespace separately
[-Wc++17-extensions]
37 | namespace QuantLib::ext {
    | ~~~~~
    | { namespace ext
1 warning generated.
In file included from TreePricerWithObserver.cpp:7:
In file included from ./TreeObservable.h:10:
In file included from /opt/homebrew/include/ql/patterns/observable.hpp:32:
In file included from /opt/homebrew/include/ql/errors.hpp:29:
/opt/homebrew/include/ql/shared_ptr.hpp:37:19: warning: nested namespace definition is a C++17 extension; define each namespace separately
[-Wc++17-extensions]
37 | namespace QuantLib::ext {
    | ~~~~~
    | { namespace ext
1 warning generated.
Arrays.cpp:247:8: warning: extra tokens at end of #endif directive [-Wextra-tokens]
247 | #endif RANGE_CHECKING
    | ^
    | //
Arrays.cpp:265:8: warning: extra tokens at end of #endif directive [-Wextra-tokens]
265 | #endif RANGE_CHECKING
    | ^
    | //
2 warnings generated.

Number of steps
50

Parameters Used:
Spot: 50
Strike: 50
Barrier: 70
Volatility: 0.3
Risk-free rate: 0.05
Dividend yield: 0.08
Expiry: 1
Number of steps: 50

Pricing Options with Observer Notifications:

CRRBiTree - 2026-05-10: American Call - CRR: 4.00451, Trinomial: 3.99863
CRRBiTree - 2026-05-10: American Put - CRR: 8.07813, Trinomial: 8.08843
CRRBiTree - 2026-05-10: Barrier Call - CRR: 1.13964, Trinomial: 1.17925
CRRBiTree - 2026-05-10: Barrier Put - CRR: 8.03444, Trinomial: 8.04969
TriTree - 2026-05-10: American Call - CRR: 4.00451, Trinomial: 3.99863
TriTree - 2026-05-10: American Put - CRR: 8.07813, Trinomial: 8.08843
TriTree - 2026-05-10: Barrier Call - CRR: 1.13964, Trinomial: 1.17925
TriTree - 2026-05-10: Barrier Put - CRR: 8.03444, Trinomial: 8.04969
```

PART 6

Warnings generated:

Parameters Used:
Spot: 50
Strike: 50
Barrier: 70
Volatility: 0.3
Risk-free rate: 0.05
Dividend yield: 0.08
Expiry: 1

Option Pricing Report via Observer Pattern

--- N = 50 ---
BiTAmerica - 2026-05-10: American Call priced at 4.00451
BiTAmerica - 2026-05-10: American Put priced at 8.07813
BiTEuroBaC - 2026-05-10: Barrier Call priced at 1.13964
BiTEuroBaP - 2026-05-10: Barrier Put priced at 8.03444

--- N = 100 ---
BiTAmerica - 2026-05-10: American Call priced at 4.00373
BiTAmerica - 2026-05-10: American Put priced at 8.08843
BiTEuroBaC - 2026-05-10: Barrier Call priced at 1.12594
BiTEuroBaP - 2026-05-10: Barrier Put priced at 8.03683

--- N = 200 ---
BiTAmerica - 2026-05-10: American Call priced at 4.00669
BiTAmerica - 2026-05-10: American Put priced at 8.08526
BiTEuroBaC - 2026-05-10: Barrier Call priced at 1.12003
BiTEuroBaP - 2026-05-10: Barrier Put priced at 8.02913

--- N = 400 ---
BiTAmerica - 2026-05-10: American Call priced at 4.00654
BiTAmerica - 2026-05-10: American Put priced at 8.08571
BiTEuroBaC - 2026-05-10: Barrier Call priced at 1.09862
BiTEuroBaP - 2026-05-10: Barrier Put priced at 8.0261